Version: V1.0.0

Confidential

Markon Electronics Corporation Private Limited

(Wetrack140 / AIS-140)

Revision History

Revision Record

Author **Author**	date **Date**	version **Version**	Check **Review**	Approve **Approve**	Description **Remarks**
Sameer	**2019-06-28**	**1.0.0**			Consolidation based on the original **Collated based on original document**
					Specification describes supplement login packet and package heartbeat reply packet
					After the start delimiter character increases comma, sos warning message header of EMR, delete login packet terminator
					Standardized description, added login packet and response packet of Heartbeat packet
					NRM packet description
					Packet Samples

Content

1. Protocol packet format 5
2. Protocol Specifications 7
2.1 Login Packet 7
2.2 Positioning/ Alert Packet 9
2.3 Device Status Info Packet (Heartbeat Packet) 17
2.4 Emergency Alert Packet (Government Server) 19
2.5 Emergency Message Format 20
Packet Samples .....

I. Protocol packet format protocol packet format

Device transmits to the server in ASCII string format data packets between each field with a comma ","
separated, beginning with "$", "*" end. format:

$, [Header], [character content] <word

Character content> <,
character content> <checksum>
* Description:

1, [character content] according to the actual filling;

2, <check> is options, log heartbeat packet and the packet does not include;

3, numerical data packet, only a check value, the base station information CELL ID and LAC represents hexadecimal, the other is a decimal.

The device sends data packets to the server in ASCII string format. The fields are separated by a comma "," and begin with "$" and end with "*".

Format:

$ [Packet heading], [Character Content], <Character Content>

<, Character Content> <, Check> * Description:

1, [character content] is filled according to the actual;

2, <check> is optional, login packet and heartbeat packet are not included;

3. For the value in the data packet, only the check value, the LAC and CELL ID in the base station information are expressed in hexadecimal notation, and the others are all decimal.

4, Emergency alarm on the package, with "*The end of the check, "the way, the customer is uploaded to the host government monitoring server (analogous to the Ministry of standard protocols), it does not explain this strange format.

4. The emergency alert packet is ended with "* check" and is uploaded to the government monitoring server of the customer's host country (similar to the Ministry standard protocol), this strange format is not explained.

Second, the protocol specification Protocol Specifications

2.1 Login Packet

$,LGN,MARK,000000000,358980100422519,V0.0.1,AIS140,26.854004,N,75.766891,E,*

Device uploads packet

Field	Description	Sample Data Example
Start Character start bit	**$**	**$**
Packet Header	**LGN**	**LGN**
Vendor ID Manufacturer ID	Manufacturer's Name Manufacture name	**MARK**
Vehicle Registration No.	**Vehicle number on which the device is installed**	**DL1PC9821**
MEI	**IMEI**	**123456789012345**
Firmware Ver. Firmware version**(four** Bit)	**Version of the firmware used in the hardware.**	**V0.0.1**
Protocol Ver. Protocol Version	**Version of the frame format protocol.**	**AIS140**
Latitude	The current setpoint value of the latitude	**12.976545**
Latitude Dir	**N / S**	**N**
Longitude	The current setpoint value longitude	**77.549759**
Longitude Dir	**E / W**	**E**
End character	*	*

Login packet response. (sample response - $LGN26032021090946*)

Field	Description	Sample Data Example
Start Character	$	$
Packet Header	LGN	LGN
Date Time	UTC (format 24 hrs)	Date value as per GPS date time per GPS time (DDMMYYYYhhmmss)
End character	*	*

2.2 Locate/Alarm data packet Positioning / Alert Packet

Sample packet - $,NRM,MARK,V0.0.1,NR,01,L,358980100422519,000000000,0,23112022,071353,26.854004,N,75.766891,E,0.0,170.70,7,376.7,0.0,0.0,Airtel,1,1,0.0,3.8,0,C,31,404,45,6237,988C,6237,4CCD,29,6237,4CCE,24,6237,C815,21,6237,988D,21,1010,00,0.2,100000,000045,0,0,0,D01B,*

Field	Data Type Data	Valid Range	Sample Data Examples	Description
Start Character	**String of characters**	**$**	**$**	**$**
Header	**String of characters**	**constant string fixed value**	**NRM**	**The header of the packet / identifier**
Vendor ID Manufacturer ID	**String of characters**	**constant string fixed value**	**MARK**	**Vendor identification header**
Firmware Version Firmware version	**String of characters**	**NA**	**V0.0.1**	**Version details of the firmware**

Packet Type Packet Type	**String of characters**	**NR / EA / TA / HP / IN / IF / BR / BD / BL / BH / HA / HB /RT / OS / OT/ GI / GO / EO**	**NR**	**NR: Normal periodic packet reports the timing data packet** **~~HP: Health packet heartbeat packet~~TA: Tamper alert dismantle alarm** **EA: Emergency alert emergency alarm** **IN: Ignition On alert ACC ON alarm IF: Ignition OFF alert ACC OFF alarm** **BR: Mains reconnected alert reconnect main power** **BD: Mains disconnected alert disconnected from the mains** **BL: Low battery alert Built-in low battery** **BH: Low battery charged alert external low voltage level** **OT: Configuration over the air alert** Alarm settings are modified **HA: Harsh acceleration alert alarm rapid acceleration** **HB: Harsh braking alert alarm brakes** **RT: Harsh / Rash turning alert alarm sharp turn** **OS: Over Speed Alert Speed alarm modifies EO: Emergency off alarm acknowledge**

				GI: into the electronic fence alarm **GO: an electronic fence alarm** **BC: Built-in low battery alert lifted**	
Alert ID Call the police **ID**	**Uint8 8 bits no break** No plastic surgery	**1 to 19** **0-255**	**01**	**Default message coming from each device** Data uploaded regularly	

			02	**Location Update (history) Would be sent, if GPRS is not available at the time of sending the message in protocol format Zero, BLANK, NIL,** **etc. make up the historical data transfer**
			03	**Alert - Disconnect from main battery If device is disconnected from vehicle battery and running on its internal battery** Disconnected from the mains
			04	**Alert - Low battery If device internal battery has fallen below a defined threshold** Built-in battery low
			05	**Alert - Low battery removed** Built-in battery low lift
			06	**Alert - Connect back to main battery Indicates that device is connected back to main battery** Main power reconnected
			07	**Alert - Ignition ON Indicates that Vehicle's Ignition is switched ON** **ACC ON alarm**
			08	**Alert - Ignition OFF Indicates that Vehicle's Ignition is switched OFF** **ACC OFF alarm**

			~~09~~	~~Alert - GPS box opened (Optional) Optional~~ message would be generated indicating GPS box opened~~d~~

			10	**Alert - Emergency state ON * When any of the emergency button is pressed** Emergency alarm
			11	**Alert - emergency State OFF When emergency state of vehicle is removed** Emergency alert lifted
			12	**Alert Over the air parameter change When any parameter is changed over the air. Shall include the name of parameter changed and source of command alarm settings are changed server1, - command send by server1 TIMER, - name of command 1 or 0 - 1 means success, 0 means fail**
			13	**Harsh Braking Alert indicating for harsh braking.** Alarm brakes
			14	**Harsh Acceleration Alert indicating for harsh acceleration.** Hard acceleration alarm
			15	**Rash Turning Alert indicating for Rash turning.** Sharp turn alarm
			~~**16**~~	~~**Alert Indicating Emergency button wire disconnect / wire cut etc. Device Tempered**~~
			17	**Overspeed**

			18	**Geo-fence Entry**
			19	**Geo-fence Exit**

			~~20~~	~~Overspeed + Geo-fenceEntry~~
			~~twenty one~~	~~Overspeed + Geo-fence Exit~~

			twenty two	**TILT rollover**
			twenty three	Wicket state
			twenty four	Into the blind area alarm
			25	NEW YORK low battery alarm
			26	Leaving the dead zone alarm
Packet Status Real-time / fill pass flag	**String** character	**H / L H complement pass / L real time**	**L**	**L = Live or H = History L = real-time reporting, H = pass up history**
IMEI	**String** character	**String of 15 characters 15 strings**	**351,732,050,525,917**	**Identified of the sending unit. 15 digit standard unique IMEI no.**
Vehicle Reg. No	**String** character	**String of 16 characters 16 strings**	**AP09BU9365**	**Mapped vehicle registration number**
GPS Fix GPS positioning status	**Uint8 8 unsigned integers**	**0/1**	**1**	**1 = GPS fix OR 0 = GPS invalid 1 = GPS positioning is not 0 = GPS positioning**
Date date	**Day: Uint8 Month: Uint8 Year: Uint16**	**Day: 1-31 Month: 1-12 Year: 2018-2999**	**15,122,017**	**Date value as per GPS date time per GPS date time (DDMMYYYY)** Date, format DDMMYYYY
Time time	**Hour: Uint8 Minute: Uint8 Second: Uint8**	**Hour: 0-23 Minute: 0-59 Second: 0-59**	**131 304**	**Time value as per GPS date time in UTC format (hhmmss)** Time, hhmmss format

Latitude	**Float**	**00.000000 - 90.000000**	**12.955446**	**Latitude value in decimal degrees**
Latitude Dir Latitudinal direction	**String** character	**N / S**	**N**	**Latitude Direction.** **N = North, S =South**

Longitude	**Float**	**00.000000 - 180.000000**	**077.637344**	**Longitude value in decimal degrees**
Longitude Dir Longitude direction	**String** character	**E / W**	**E**	**Longitude Direction.** **E = East, W = West**
Speed speed	**Float** Float	**000.0 to 550.0**	**000.1**	**Speed of Vehicle as Calculated by GPS module in** **VLT.(in km / hrs.)** unit **KM / H**
Heading course	**float** Float	**000.00 to 359.99**	**000.00**	**Course over ground in degrees**
No of Satellites Satellites	**Uint8** **8 unsigned integers**	**0-24**	**10**	**Number of satellites available for fix**
Altitude altitude	**float** Float	**-1000.0 to18000 meters** **-1000.0 to 18,000 meters**	**00,913.2**	**Altitude of the device in meters**
PDOP	**float**	**000.0 to 099.9**	**001.6**	**Positional dilution of precision**
HDOP	**float**	**000.0 to 099.9**	**000.8**	**Horizontal dilution of precision**
Operator Name Carrier Name	**String** character	**String of maximum 11 characters** String, maximum 11	**VODAFONE IN**	**Name of the operator** Carrier Name
Ignition **ACC status**	**Uint8 8 bits no break** No plastic surgery	**0/1**	**1**	**1 = Ignition On, 0 = Ignition Off** **1 = ACC ON 0 = ACC OFF**

Main Power Status Main power state	**Bool** Boolean value	**0/1**	**1**	**0 = Vehicle Battery disconnected** **0 = disconnected from the mains** **1 = Vehicle Battery reconnected** **Reconnect main power source 1 =**
Main Input Voltage Mains voltage	**float** Float	**7.0-40.0**	**11.9**	**Indicator showing source voltage in Volts.** Mains voltage in V

Internal Battery Voltage Built-in battery voltage	**float**	**0.0-5.8**	**05.5**	**Indicator for level of battery charge remaining.** Built-in battery voltage, in V
Emergency Status Emergency sign	**Uint8 8 bits no break** No plastic surgery	**0/1**	**0**	**1 = On, 0 = Off**
Tamper Alert Dismantle alarm (optional)	**String** character	**O / C**	**C**	**C = Cover Closed, Closing the lid** **O = Cover Open the cover open**
GSM Signal Strength **GSM signal strength**	**Uint8 8 bits no break** No plastic surgery	**0-31**	**twenty one**	**Value Ranging from 0 - 31** The signal strength, range 0-31
MCC	**String of characters**	**000-999**	**404**	**Mobile Country Code**
MNC	**String of characters**	**000-999**	**86**	**Mobile Network Code**
LAC	**String of characters**	**0000-FFFF**	**797F**	**Cell 0 LAC Location Area Code**
Cell ID	**String of characters**	**0000-FFFF**	**4508**	**GSM Cell ID cell 0 Cell ID**
NMR1 LAC	**String of characters**	**0000-FFFF**	**797F**	**Neighboring Location Area Code 1 LAC cell**
NMR1 Cell Id	**String of characters**	**0000-FFFF**	**4509**	**Neighboring GSM Cell ID cell 1 Cell ID**
NMR1 Signal strength	**String of**	**0-31**	**twenty two**	**Neighboring Signal strength the signal**

	characters			strength of a cell
NMR2 LAC	String of characters	0000-FFFF	C363	Neighboring Location Area Code cell 2 LAC
NMR2 Cell Id	String of characters	0000-FFFF	1287	Neighboring GSM Cell ID cell 2 Cell ID
NMR2 Signal strength	String of characters	0-31	11	Neighboring Signalstrengthcellsignal strength 2
NMR3 LAC	String of characters	0000-FFFF	797F	Neighboring Location Area Code 3 LAC cell
NMR3 Cell Id	String of characters	0000-FFFF	4507	Neighboring GSM Cell ID cell 3 Cell ID
NMR3 Signal strength	String of characters	0-31	10	Neighboring Signal strength signal strength of the cell 3
NMR4 LAC	String of characters	0000-FFFF	797F	Neighboring Location Area Code cell 4LAC
NMR4 Cell Id	String of characters	0000-FFFF	4507	Neighboring GSM Cell ID cell 4 Cell ID
NMR4 Signal strength	String of characters	0-31	05	Neighboring Signalstrengthcellsignal strength 4

Digital Input Status Digital input status	**DIN1:** **bool** **DIN2:** **bool** **DIN3:** **bool** **DIN4:** **bool** Boolean value	**DIN1:** **0/1** **DIN2:** **0/1** **DIN3:** **0/1** **DIN4:** **0/1**	**0000**	**4 external digital input status (Status of Input 1 to Input 3 (0 = Off; 1 = On))** Digital input status **0 0 = Off, 1 = On** **Remark – i) for acc** **ii) for sos** **iii) for ac or door** **iv) empty**
Digital Output Status Digital output state	**D01: bool** **Boolean value** **D02: bool** Boolean value	**D01: 0/1** **D02: 0/1**	**00**	**2 external digital output status (0 = Off; 1 = On)** Digital output state **0 = Off, 1 = On** **i) Relay** **ii) TTL(ex- rfid)**
Analog Input Analog Input Voltage	**Float**	**0-30.0 V**	**12.9V**	External voltage input
Delta Distance Mileage statistics	**Int integer**	**0-4294967295** (0xFFFFFFFF)	**1000000**	Unit: m
Frame Number serial number	**Uint32 32 bit** **no break** No plastic surgery	**000001--999999**	**000 580**	**Sequence Number of the messages (000001 to 999999)** Serial number range**000001-999999**

OTA Response format	string	*	*	**Server1 – command send by server1 TIMER - name of command 1 – status of command 0-failed, 1- success**	
OTA Response Modify settings command source	String	:(instruction format source, referring to Order ID, status) E.g (USB/SMS/SERVER1/ SERVER2, CFG_HBT, 0 = F **ail 1 = Success)**		**(SERVER1, timer, 1)**	
Checksum Check Digit	**String of characters**	**0000 - FFFF**	**EDEA**	**Insures No error in transmission** Parity bits, CRC from the header to the serial number range	
End character Terminator	**String** character	*	*	*	

Description:

1. For the other packet type is added, can be added on their own, but can not duplicate existing packet type.

2, For other alarm ID is added, can be added on their own, but can not duplicate existing alarm ID.

3, the base station 0 to populate the missing information fields.

Description:

1. You can add other packet types, but the packet added should be different from existing packets.

2. You can add other alert ID, but the ID added should be different from existing ID ..

3. When the base station information is missing, the corresponding field is filled with 0.

2.3 Device status information packet (heartbeat packets)

Device Status Info Packet (Heartbeat Packet)

Uploading data packet

Data packet uploaded by device.

Sample packet - $,HBT,MARK,V0.0.1,358980100422519,31,20,0,10,10,1010,0.0,0.2*

Field	Description	Sample Data
Start Character Start character	**Start character**	**$**
Packet Header	**HBT**	**HBT**
Vendor ID name of the manufacturer		**MARK**
Firmware Version Firmware version	Firmware version of the device currently being used	**V1.0.0**
IMEI	Device IMEI number	**351,732,050,525,917**
Internal Battery Percentage Built-in battery percentage	Built-in battery percentage	**65**
Low battery threshold Percentage Low battery alarm threshold percentage	Low battery alarm threshold percentage	**20**
Memory percentage Data Memory usage percentage	**0.33**	**Indicates flash memory used in percentage**
Data update rate when ignition ON **ACC ON data upload interval**	**ACC ON data upload interval (s)**	**10**
Data update rate when ignition OFF **ACC OFF data upload interval**	**ACC OFF data upload interval (s)**	**60**
Digital Input status Digital input status	Digital input status	**0001 (DIN1 = 0, DIN2 = 0, DIN3 = 0, DIN4 = 1)**
Analog Input 1	Analog input status (in V)	**0.0**

Analog input 2	Analog input status (in V)	**0.0**
End Character terminator	**End character**	*

The server returns the packet

Server Response

Field	Description	Sample Data Example
Start Character start bit	**$**	**$**
Packet Header	**HBT**	**HBT**
End character	*	*

2.4 Emergency alarm package (government server) Emergency Alert Packet (Government Server)

Field	Sample Data	Description
Start Character start symbol	**$**	**Start character**
Packet Header	**EPB**	**EPB, The unique identifier for all messages from VLT** **EPB, a fixed value**
~~**Vendor ID name of the manufacturer**~~	~~**MARK**~~	
Message Type Information Type	**EMR**	**Message Types supported. Emergency Message (EMR) or Stop Message (SEM)** **EMR , SEM =**
IMEI	**123456789012345**	**Identified of the sending unit. 15 digit standard unique IMEI no.**
Packet Type Real-time / fill pass flag	**NM**	**NM - Normal Packet, real-time upload** **SP - Stored Packet, fill biography**
Date Time Date Time	**18122017124850**	**Date value as per GPS date time per GPS time (DDMMYYYYhhmmss)** Date and time, format DDMMYYYYhhmmss
GPS Fix **GPS positioning status**	**A**	**A = GPS fix OR V = GPS invalid** **A = GPS positioning has V = GPS positioning is not**
Latitude	**12.955451**	**Latitude value in decimal degrees**

Latitude Dir Latitudinal direction	**N**	**Latitude Direction.** **N = North, S = South**
Longitude	**077.637313**	**Longitude value in decimal degrees**
Longitude Dir	**E**	**Longitude Direction.** **E = East, W = West**
Altitude above sea level	**00,908.0**	**Altitude of the device in meters unit of M**
Speed	**000.0**	**Speed of Vehicle as Calculated by GPS module in VLT. (In km/ hrs.) Units KM/H**
Distance distance	**0000.0**	**Distance calculated from previous GPS data. Calculation of distance starts once SOS is pressed and cleared once SOS is cleared** The current anchor point a distance from the anchor point on
Provider Positioning Type	**G**	**G - Fine GPS** **N - Coarse GPS or data from the network** **N = G = GPS positioning base station location**
Vehicle Reg. No number plate	**AP09BU9365**	**Mapped vehicle registration number**
Reply Number Receiving alarm SMS numbers	**NA**	**The mobile number to which Test response needs to be sent** Here to receive alarm SMS mobile phone number if no number is null
End Character Terminator	*	**End character**
Checksum Check Digit	**FFFFFF96**	**The 32 bit checksum of all the characters from the header up to the CRC field** **32bit CRC, CRC ranging from the header to the terminator**

2.5 Emergency alarm SMS format **Emergency Message Format**

Field	Description	Sample Data
Alert No		**EMR**
IMEI	**Identified of the sending unit. 15 digit standard unique IMEI no.**	**123456789012345**
Latitude	**Latitude value in decimal degrees**	**12.955451**
Latitude Dir Latitudinal direction	**Latitude Direction.** **N = North, S = South**	**N**
Longitude **Longitude**	**Longitude value in decimal degrees**	**077.637313**
Longitude	**Longitude Direction.** **E = East, W = West**	**E**
GPS Fix **GPS positioning status**	**A = GPS fix OR V = GPS invalid** **A = GPS positioning has V = GPS positioning is not**	**A**
Speed	**Speed of Vehicle as Calculated by GPS module in VLT. (In km / hrs.)** **Units KM /H**	**000.0**
MCC	**Mobile Country Code**	**404**
MNC	**Mobile Network Code**	**86**
LAC	**Cell 0 LAC Location Area Code**	**797F**
Cell ID	**GSM Cell ID cell 0 Cell ID**	**4508**
Date Time		**DD / MM / YYYY HH: MM: SS**

Packet Sample -

(Live packet)

$,NRM,MARK,V0.0.1,NR,01,L,358980100675496,DL17S4979,1,21032023,100451,28.360100,N,76.927536,E,0.4,68.02,10,258.2,1.5,1.3,Airtel,1,1,17.2,4.0,1,C,18,404,10,209,3185,209,6D17,23,209,7510,17,209,C176,17,209,6D18,17,1110,00,0.1,2023,000160,0,0,0,4D17,*

(History packet)

$,NRM,MARK,V0.0.1,NR,02,H,358980100675496,DL16P5555,1,21032023,142106,28.360180,N,76.927727,E,0.3,273.23,13,276.3,1.8,1.1,Airtel,1,1,17.4,4.0,0,C,18,404,10,209,6D18,209,6D17,21,209,4333,20,209,B2F8,19,0,0,0,1110,00,0.1,2023,000569,0,0,0,CB0C,*

(Battery Disconnect alert packet)

$,NRM,MARK,V0.0.1,BD,03,L,358980100675496,DL17S4979,1,21032023,095749,28.360096,N,76.927513,E,0.6,97.56,10,259.7,1.5,1.3,Airtel,0,0,0.0,4.0,0,C,17,404,10,209,D2FC,209,6D17,26,209,B2F8,20,209,C176,19,0,0,0,0110,00,0.1,2023,000113,0,0,0,6F49,*

(Battery low alert packet)

$,NRM,MARK,V0.0.1,BL,04,L,358980100675496,DL17S4979,0,21032023,101800,28.360083,N,76.927666,E,0.0,230.91,9,290.6,0.0,0.0,Airtel,0,0,8.9,3.5,0,C,12,404,10,209,B2F8,209,6D17,16,209,754,14,209,C176,13,209,C343,12,1110,00,0.1,24,000008,0,0,0,2F7C,*

(Low battery remove alert packet)

$,NRM,MARK,V0.0.1,BC,05,L,358980100675496,DL17S4979,0,21032023,103049,28.360083,N,76.927666,E,0.0,230.91,9,290.6,0.0,0.0,Airtel,1,1,8.9,3.9,0,C,12,404,10,209,B2F8,209,6D17,16,209,754,14,209,C176,13,209,C343,12,1110,00,0.1,24,000008,0,0,0,2F7C,*

(Battery reconnect alert packet)

$,NRM,MARK,V0.0.1,BR,06,L,358980100675496,DL17S4979,1,21032023,095758,28.360096,N,76.927513,E,0.8,97.56,10,259.5,1.5,1.3,Airtel,0,1,0.0,4.0,0,C,17,404,10,209,B2F8,209,C176,19,0,0,0,0,0,0,0,0,0,0110,00,0.1,2023,000116,0,0,0,D6D3,*

(ignition on alert packet)

$,NRM,MARK,V0.0.1,IN,07,L,358980100675496,DL17S4979,1,21032023,095811,28.360064,N,76.927567,E,1.1,144.87,10,260.2,1.5,1.3,Airtel,1,1,17.2,4.0,0,C,19,404,10,209,B2F8,209,6D17,24,209,C176,18,0,0,0,0,0,0,1110,00,0.1,2023,000118,0,0,0,63F0,*

(ignition off alert packet)

$,NRM,MARK,V0.0.1,IF,08,L,358980100675496,DL17S4979,1,21032023,095757,28.360100,N,76.927513,E,0.7,97.56,10,259.5,1.5,1.3,Airtel,0,0,0.0,4.0,0,C,17,404,10,209,B2F8,209,C176,19,0,0,0,0,0,0,0,0,0,0110,00,0.1,2023,000114,0,0,0,D965,*

(emergency on alert packet)

$,NRM,MARK,V0.0.1,EA,10,L,358980100675496,DL17S4979,1,21032023,100422,28.360107,N,76.927521,E,0.8,68.02,11,258.1,1.5,1.3,Airtel,1,1,17.3,4.0,1,C,19,404,10,209,3185,209,6D17,21,209,7510,18,209,C176,17,209,B2F8,17,1110,00,0.1,2023,000157,0,0,0,0440,*

(emergency off alert packet)

$,NRM,MARK,V0.0.1,EO,11,L,358980100675496,DL17S4979,1,21032023,095433,28.360119,N,76.927521,E,1.1,40.08,9,260.2,1.8,1.5,Airtel,1,1,17.3,4.0,0,C,17,404,10,209,38B5,0,0,0,0,0,0,0,0,0,0,0,0,1110,00,0.1,2023,000083,0,0,0,D602,*

(Over the air parameter change alert packet)

$,NRM,MARK,V0.0.1,OT,12,L,358980100675496,DL17S4979,1,21032023,101659,28.360104,N,76.927467,E,0.4,355.41,11,258.6,1.6,1.3,Airtel,1,1,17.3,4.0,1,C,19,404,10,209,3185,0,0,0,0,0,0,0,0,0,0,0,0,1110,00,0.1,2023,000233,SMS,917011440510,SOESERVER,1,11AA,*

(Harsh break alert packet)

$,NRM,MARK,V0.0.1,HB,13,L,358980100675496,DL17S4979,1,21032023,100441,28.360104,N,76.927528,E,0.2,68.02,11,258.1,1.5,1.3,Airtel,1,1,17.3,4.0,1,C,19,404,10,209,3185,209,6D17,22,209,6D18,17,209,7510,17,209,B2F8,17,1110,00,0.1,2023,000159,0,0,0,F406,*

(Harsh acceleration alert packet)

$,NRM,MARK,V0.0.1,HA,14,L,358980100675496,DL17S4979,1,21032023,100431,28.360104,N,76.927528,E,1.8,68.02,11,258.1,1.5,1.3,Airtel,1,1,17.3,4.0,1,C,19,404,10,209,3185,209,6D17,22,209,7510,17,209,B2F8,17,209,C176,17,1110,00,0.1,2023,000158,0,0,0,A9B9,*

(Rash Turn alert packet)

$,NRM,MARK,V0.0.1,RT,15,L,358980100675496,DL17S4979,1,21032023,100501,28.360100,N,76.927536,E,1.5,68.02,11,258.3,1.5,1.3,Airtel,1,1,17.3,4.0,1,C,20,404,10,209,3185,209,6D17,22,209,6D18,17,209,C176,17,209,7510,17,1110,00,0.1,2023,000161,0,0,0,E049,*

(Device Tampered alert packet)

$,NRM,MARK,V0.0.1,DT,16,L,358980100675496,DL17S4979,1,21032023,101710,28.360104,N,76.927460,E,0.0,355.41,11,258.6,1.6,1.3,Airtel,1,1,17.3,4.0,0,C,19,404,10,209,3185,209,6D17,18,209,7510,17,0,0,0,0,0,0,1010,00,0.1,2023,000235,0,0,0,6CA0,*

(Login packet sample)

$,LGN,MARK,000000000,358980100422519,V0.0.1,AIS140,26.854004,N,75.766891,E,*

(Heartbeat packet sample)

$,HBT,MARK,V0.0.1,358980100422519,31,20,0,10,10,1010,0.0,0.2*